

Switchable Single-Axis Preference Auditors over a Draft Model

A draft–auditor decoding scheme with unmerged DPO adapters, two acceptance regimes, and a diagnostic for when pairwise selection beats argmax

Abstract

We study a decoding-time alignment scheme in which a fast *draft* model proposes text and a single, hot-swappable *Direct Preference Optimization (DPO) adapter*—a “blade”—audits it along one alignment axis at a time. Because the adapter is kept *unmerged*, its DPO implicit reward is computed from one model as the log-probability gap between the adapter-enabled and adapter-disabled passes, requiring no separate reward model. We make explicit two ways to use the blade: (i) as a *scalar gate* that tilts generation toward its axis (in both a threshold variant and a principled tilted-policy variant), and (ii) as a *joint-pairwise judge* that compares whole candidates and feeds a tournament. We show that the scalar mode is transitive—so a tournament over it provably reduces to argmax—and that the pairwise mode is the only one that can beat argmax, which it does only when its verdicts are empirically non-transitive. We position the scheme against decoding-time multi-objective alignment (MOD) and show where it genuinely differs and where it collapses to it. Evaluation combines behavioral proxies (G-Eval, toxicity) with the intrinsic, latent-geometry Alignment Quality Index (AQI) on the LITMUS dataset.

1 Thesis and contributions

A single unmerged DPO adapter on a frozen base model can audit a *separate* draft model along one preference axis (e.g. helpfulness, harmlessness, truthfulness), and the active axis can be hot-swapped at runtime. The adapter exposes two distinct objects: a *scalar implicit reward* and an *implicit pairwise comparator*. The central scientific question is *when each should drive decoding*, and in particular when a pairwise tournament does anything that scalar argmax does not.

Contributions.

1. A decoupled, hot-swappable single-auditor decoding architecture (frozen draft + unmerged DPO blade), with the implicit reward obtained from adapter-on/adapter-off passes of one model.
2. Two acceptance regimes for the gate—an RSD-style threshold and a GSI-style principled tilted-policy approximation—and an analysis of their targets.
3. A *one-blade, two-roles* treatment (scalar gate vs. pairwise judge) and a per-axis diagnostic predicting when pairwise tournament selection beats argmax.
4. An explicit account of how the scheme differs from, and where it degenerates to, decoding-time multi-objective alignment (MOD).

2 Architecture

2.1 The implicit-reward auditor without a separate reference

Each blade is a LoRA adapter trained with DPO. By the DPO identity, the adapter encodes an implicit reward $r(y | x) = \beta \log(\pi_{\text{blade}}(y | x) / \pi_{\text{ref}}(y | x))$. Keeping the adapter *unmerged* makes

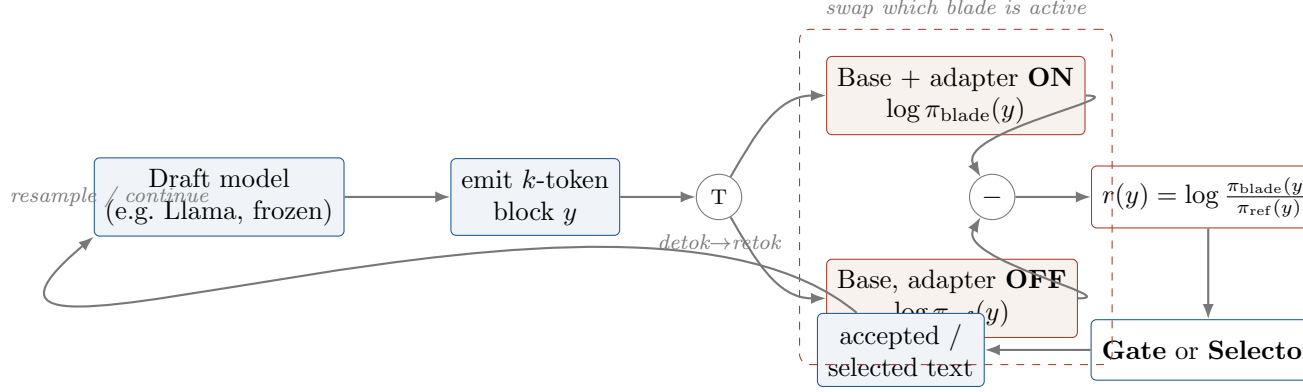


Figure 1: The draft model proposes a block of text; the auditor scores it as the DPO implicit reward, computed as the difference between adapter-on and adapter-off passes of a single frozen base. Different draft and base models mean scoring happens on *text* (detokenize then re-tokenize), not on token IDs. The active blade is swapped at runtime.

both terms available from one model: enabling the adapter gives π_{blade} , disabling it gives π_{ref} . The reward of a block is therefore

$$r(y) = \log p_{\text{base}}(y \mid \text{adapter on}) - \log p_{\text{base}}(y \mid \text{adapter off}),$$

with no separate reward model loaded. Merging the adapter would destroy π_{ref} and with it the reward; the unmerged form is required, not merely convenient. The adapter-off pass can be cached and shared across candidates and across blades, roughly halving auditor cost.

2.2 Cross-tokenizer, block-level operation

When the draft and the auditor base come from different model families, their vocabularies differ, so token-level verification is impossible and no speedup of classical speculative decoding is available. The auditor therefore scores *text*: the draft’s block is detokenized to a string and re-tokenized under the auditor’s tokenizer before scoring. This makes the scheme an *alignment* mechanism (it intentionally changes the output distribution), explicitly distinct from speculative decoding for speed (which preserves the distribution exactly).

3 Two decoding regimes

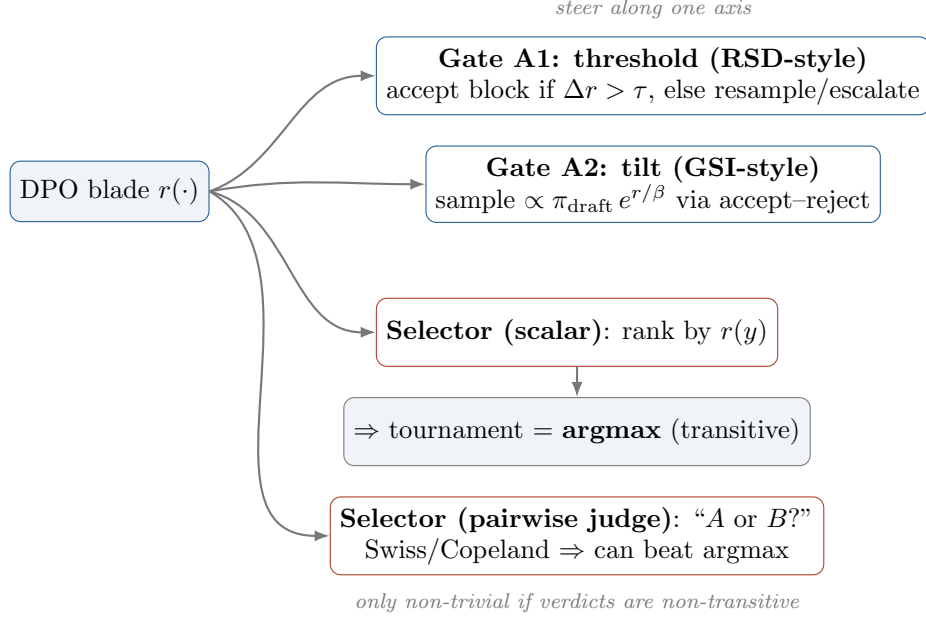
3.1 Gate A1 — threshold (RSD-style)

Blockwise, the draft proposes a k -token block; accept it if its reward improvement over the reference exceeds a threshold τ , otherwise resample (up to m tries) or escalate to a larger target model. This is simple and cheap and introduces a controlled bias toward the axis, but carries no distributional guarantee.

3.2 Gate A2 — principled tilt (GSI-style)

The reward-tilted target is

$$\pi^*(y \mid x) \propto \pi_{\text{draft}}(y \mid x) \exp(r(y \mid x)/\beta).$$



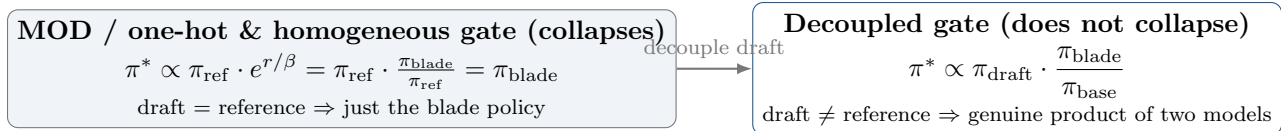


Figure 3: If the draft equals the blade’s reference (the homogeneous case, and what MOD with a one-hot weight reduces to), tilting the reference by the blade’s own implicit reward recovers the blade policy exactly—no novelty. Using a *different* draft makes the target a true product of two distinct models, which is not anything MOD produces.

that does not exist here. (3) *The orthogonal part.* MOD has no selection step, no pairwise judging, no tournament, and no diagnostic; the selector arm and the “when does pairwise beat argmax” question have no MOD counterpart at all. The scheme’s distance from MOD therefore lives in the *decoupled draft* and the *selection diagnostic*, not in single-axis gate quality.

5 Which blades to use (availability)

Off-the-shelf, single-axis preference adapters are unevenly available. Helpfulness and harmlessness/safety are well covered (Anthropic HH-RLHF and PKU-SafeRLHF preference data; PKU-Alignment’s *Beaver* ships a separate helpfulness reward and a harmlessness “cost” model; alignment-handbook publishes DPO LoRAs such as the Zephyr line). Honesty and other axes are thinner and usually trained on demand, which is cheap given existing preference datasets. A recent preference-vector study aligns along Helpful, Harmless, Honesty, and Psychocounsel simultaneously, indicating these axes are individually tractable. Table 1 summarizes practical choices; for any axis without a ready adapter, a LoRA DPO run on the listed dataset produces a blade on the same shared base.

Axis	Off-the-shelf source	Else: DPO-train on
Helpfulness	Beaver reward; alignment-handbook DPO LoRAs	UltraFeedback, HelpSteer (helpfulness)
Harmlessness / safety	Beaver “cost”; SafeRLHF-tuned models	PKU-SafeRLHF, Anthropic HH-harmless
Honesty / truthfulness	sparse; usually trained	TruthfulQA-style pairs, HelpSteer (correctness)
Toxicity (anti-)	DExperts-style experts	Jigsaw / RealToxicityPrompts-derived pairs
Conciseness / verbosity	via HelpSteer attribute labels	HelpSteer (verbosity)
Domain (e.g. counseling)	preference-vector precedent	domain preference pairs

Table 1: Single-axis blade options. All blades should be DPO-trained from the *same* base checkpoint so their implicit rewards are comparable and so adapter swapping is meaningful.

6 Evaluation

Evaluation pairs behavioral, output-level proxies with an intrinsic, representation-level metric, mirroring the framework that introduced the latter.

G-Eval. LLM-as-judge scoring (chain-of-thought, strong held-out judge) of per-axis quality and of head-to-head win rates between configurations. Used for helpfulness/instruction-following and as the gold judge in oracle-gain measurements.

Toxicity. A toxicity classifier on generations for the harmlessness axis (alongside refusal and over-refusal checks), capturing output-level safety.

AQI (Alignment Quality Index) on LITMUS. A geometric, prompt-invariant metric that measures the separation of safe vs. unsafe activations in latent space using clustering indices (Davies–Bouldin, Dunn, Xie–Beni, Calinski–Harabasz). AQI is decoding-invariant and is designed to flag hidden misalignment and “alignment faking” that behavioral proxies miss; LITMUS is its companion dataset spanning DPO/RLHF/GRPO regimes. We use AQI as the *alignment-quality index* for the audited system and report it next to G-Eval and toxicity. Because AQI is computed from internal activations, in the heterogeneous setting it is measured on a fixed probe model’s representations of the system’s outputs (stated explicitly so the metric is well defined).

Diagnostic metrics (selector arm). Per axis, over N candidates: directed 3-cycle rate, Condorcet-winner existence, position-flip rate, argmax-agreement (fraction where the Copeland winner equals the scalar-argmax winner), and oracle-gain recovery under the G-Eval judge for {argmax, soft-best-of- n , single-elim, Swiss+Copeland, round-robin+Bradley–Terry}. The headline result is a per-axis table; the tournament is justified exactly where cycle rate is non-trivial and argmax-agreement is low.

Metric	Level	What it tells us
G-Eval	behavioral (output)	per-axis quality; win rates; gold judge
Toxicity	behavioral (output)	harmlessness; over-refusal via XSTest-style probes
AQI on LITMUS	intrinsic (latent)	true alignment separation; alignment-faking / jailbreak risk
Cycle / flip / argmax-agree	selector diagnostic	whether pairwise selection can beat argmax
Oracle-gain recovery	selector diagnostic	how much best-of- N gain each rule captures
Cost: judge-calls, tokens/s	systems	gate $O(N)$, Swiss $O(N \log N)$, round-robin $O(N^2)$

Table 2: Evaluation suite: behavioral proxies, the intrinsic alignment-quality index, the selector diagnostic, and cost.

7 Experiments, baselines, and expected outcome

Research questions. (RQ1) Can the block-level gate steer a frozen draft along its axis, and at what quality/latency cost? (RQ2) Is the single-axis blade, as a pairwise judge, non-transitive enough that tournament selection beats argmax, and for which axes? (RQ3) Does hot-swapping the active blade produce a clean behavioral change without an off-axis tax, and how does single-active switching compare to weight-merged composition?

Baselines. Ungated draft; *blade-as-policy* (run the blade directly—essential given Section 4); RSD/GSI with a scalar reward model; a single-expert decode-time baseline; MOD/weight-merging for the switchability comparison; argmax-best-of- N and soft-best-of- n for selection.

Sequencing (go/no-go). (1) Confirm the blade still discriminates on draft-generated (out-of-distribution) text. (2) On harmlessness only, measure cycle rate and argmax-agreement: if cycles are negligible, ship a gate-only study; otherwise build the full selector arm. (3) Gate + switchability (RQ1, RQ3). (4) Selector + diagnostics (RQ2). (5) Cost and write-up.

Expected outcome. The gate reliably improves the active axis and switches cheaply; the tournament likely beats argmax on subjective axes (e.g. harmlessness) and collapses to argmax on factual ones (e.g. truthfulness). The deliverable is the diagnostic that predicts which case obtains, plus the decoupled switchable-auditor architecture—not a leaderboard win on single-axis quality.

Notes on related work to differentiate against

Decoding-time guidance and tilting: guided speculative inference; reward-guided speculative decoding; reward-augmented and reward-guided token decoding; controlled decoding with prefix scorers; product-of-experts decoding. Multi-objective and composable alignment: decoding-time multi-objective alignment (MOD); multi-objective DPO; weight-merging (rewarded/personalized soups); preference-vector and directional methods. Selection and preferences: knockout-tournament pairwise reward models; Nash learning from human feedback and self-play preference optimization (the theory of when pairwise preferences, not scalar rewards, are the right object). Evaluation: G-Eval; the Alignment Quality Index and LITMUS. Classical speculative decoding is cited only to contrast: it preserves the distribution, whereas this scheme intentionally tilts it.